Revision · Enlargement

Cambridge insert: the transformation that does change size — and the scale factor that measures it.

LESSON 64 | 1 × 40 MIN | GEOMETRY 8 — REVISION | STAGE 9 · 13.4

LESSON CLOCK

5' | 14' | 12' | 5' | 4'

Enlarge a triangle from a point	5 min
Scale factor	14 min
Finding the centre	12 min
Areas	5 min
Homework	4 min

LEARNING OBJECTIVES

- Enlarge a shape from a given centre by a given scale factor.
- Find the centre and the scale factor of a given enlargement.
- Use the fact that lengths scale by k and areas by k^2.
- Work with fractional and negative scale factors.

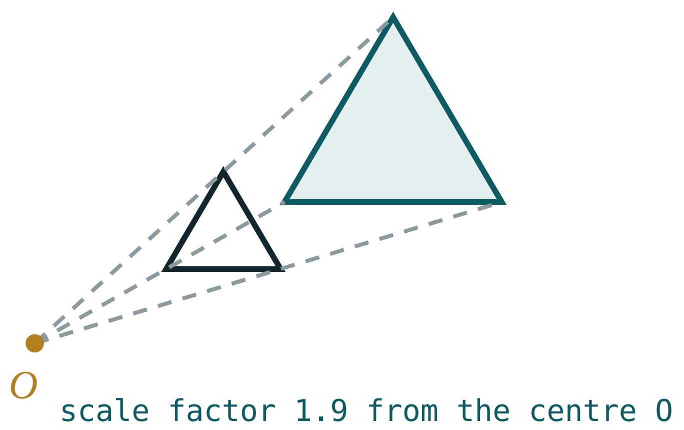
TEXTBOOK

National	Annual revision
Cambridge	Learner's Book pp. 307–314
Workbook	Workbook 13.4

01 Explanation

How an enlargement works

Choose a **centre** O and a **scale factor** k. Each point A moves along the ray OA to the point A' with $OA' = k \cdot OA$.



Rays from the centre through each vertex; the image sits k times as far along each ray.

The image is **similar** to the object: the same shape, all angles unchanged, every length multiplied by k. Only the centre stays where it is.

K	EFFECT
$k > 1$	larger, same way up
$0 < k < 1$	smaller, same way up
$k = 1$	unchanged
$k < 0$	on the other side of the centre, turned through $180°$

From the origin the rule is simply $(x; y) \to (kx; ky)$. From another centre $(a; b)$, measure from there: $(x; y) \to (a + k(x - a); b + k(y - b))$.

Finding it, and what happens to area

To find the **scale factor**: divide any image length by the matching object length. To find the **centre**: join each point to its image and extend — all the lines meet at the centre.

LENGTH, AREA AND VOLUME

$$\text{lengths } \times k,\ \ \text{areas } \times k^2,\ \ \text{volumes } \times k^3$$

An enlargement with $k = 3$ makes a shape 3 times as wide and 3 times as tall, so **9 times** the area — the same squaring rule as the unit conversions of the last lesson.

ENLARGEMENT IS NOT AN ISOMETRY

Unless $k = \pm 1$, lengths change, so the image is **similar** to the object but not congruent. Angles, however, never change.

02 Worked examples

EXAMPLE 1 **Enlarge the triangle $(1; 1)$, $(3; 1)$, $(1; 4)$ by scale factor 2 about the origin. Find the new area.**

1. $(1; 1) \to (2; 2)$

   Double each coordinate.

2. $(3; 1) \to (6; 2)$

3. $(1; 4) \to (2; 8)$

4. Original area: $\frac{1}{2} \cdot 2 \cdot 3 = 3$

5. New area: $3 \cdot 2^2 = 12$

   Areas scale by k^2.

ANSWER

$(2;2)$, $(6;2)$, $(2;8)$; area 12

EXAMPLE 2 **A shape is enlarged from $(1; 1)$ with $k = 3$. Find the image of $(4; 2)$.**

1. $(4 - 1; 2 - 1) = (3; 1)$

   The vector from the centre.

2. $3 \cdot (3; 1) = (9; 3)$

   Multiply by the scale factor.

3. $(1; 1) + (9; 3) = (10; 4)$

   Measure back from the centre.

ANSWER

$(10; 4)$

EXAMPLE 3

Two similar rectangles have corresponding sides *4 cm* and *10 cm*. The smaller has area $20\ cm^2$. Find the larger.

1. $k = \dfrac{10}{4} = 2.5$

   The scale factor.

2. $k^2 = 6.25$

   The area factor.

3. $20 \cdot 6.25 = 125\ cm^2$

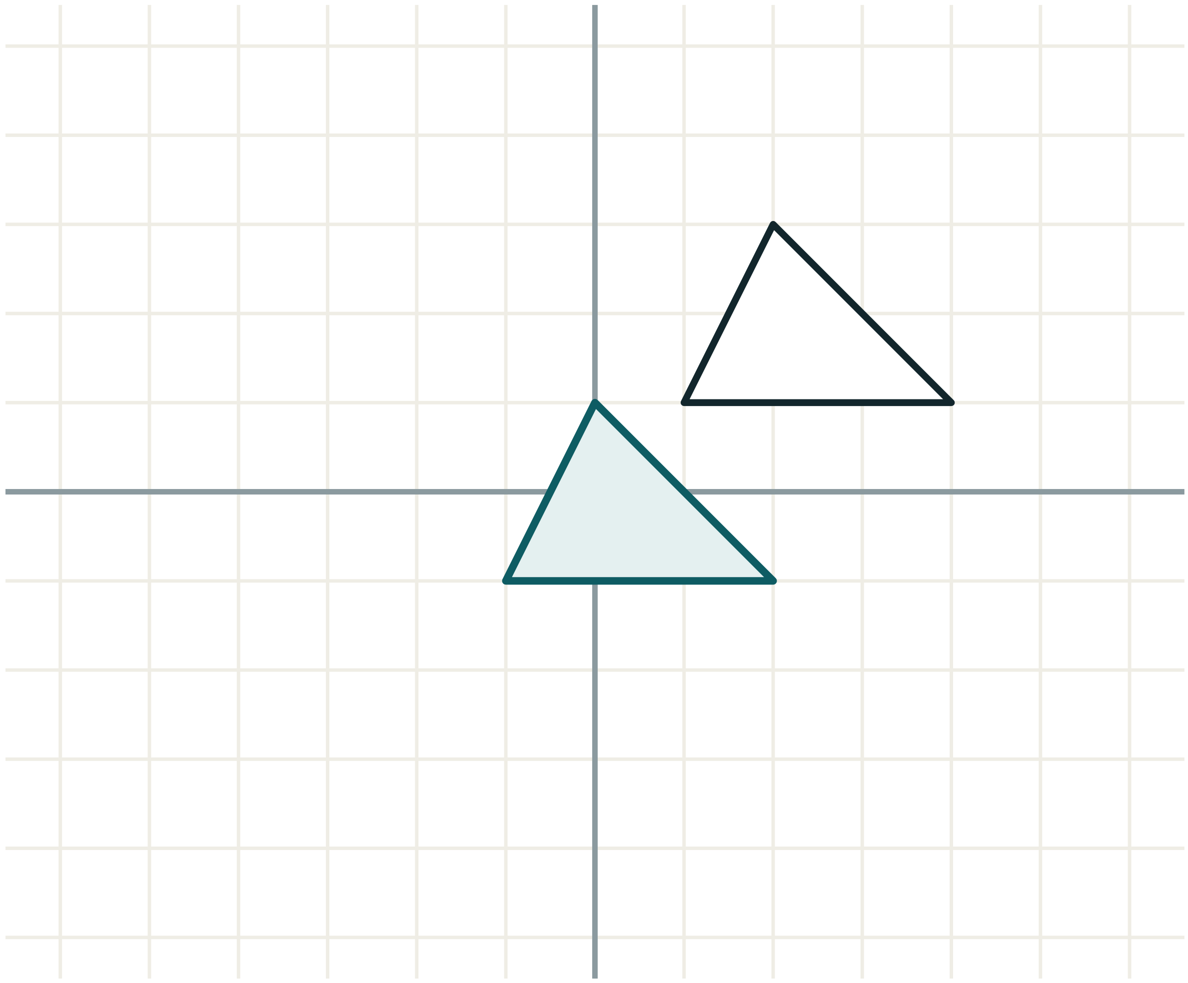
ANSWER

$125\ cm^2$

03 Interactive model

Change the scale factor and watch how far along each ray the image travels.

Enlargement from a centre



transformation **translation** | lengths **unchanged** | angles **unchanged** | orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Enlargement	O‘xshash almashtirish	Гомотетия (подобие)
Scale factor	O‘xshashlik koeffitsienti	Коэффициент подобия
Centre of enlargement	O‘xshashlik markazi	Центр гомотетии
Similar figures	O‘xshash figuralar	Подобные фигуры
Corresponding sides	Mos tomonlar	Соответственные стороны
Ray from the centre	Markazdan nur	Луч из центра
Fractional scale factor	Kasr koeffitsient	Дробный коэффициент
Negative scale factor	Manfiy koeffitsient	Отрицательный коэффициент

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Enlarging $(3; 4)$ from the origin by $k = 2$ gives:

- $(5; 6)$
- $(6; 8)$
- $(1.5; 2)$
- $(6; 4)$

If $k = 4$, areas are multiplied by:

- 4
- 8
- 16
- 64

An enlargement with $k = \frac{1}{2}$ makes the image:

- larger
- smaller
- the same size
- upside down

Under an enlargement, angles:

- double
- halve
- stay the same
- depend on k

To find the centre of an enlargement:

- take the midpoint
- join each point to its image and extend
- use the origin
- measure an angle

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Enlarge (2; 3) from the origin by $k = 3$.*

   ANSWER *(6; 9)*

2. *Enlarge (4; 8) from the origin by $k = 0.5$.*

   ANSWER *(2; 4)*

3. *If $k = 3$, by what factor does area grow?*

   ANSWER *9*

4. *If $k = 5$, by what factor does volume grow?*

   ANSWER *125*

5. *Enlarge (1; −2) from the origin by $k = 4$.*

   ANSWER *(4; −8)*

6. *Two similar shapes have sides 3 and 12. Find k.*

   ANSWER *4*

7. *A shape of area 5 is enlarged by $k = 2$. Find the new area.*

   ANSWER *20*

Medium · the standard question

7 PROBLEMS

1. *Enlarge* $(5;3)$ *from* $(1;1)$ *by* $k=2$.

   ANSWER $(9;5)$

2. *Enlarge* $(7;4)$ *from* $(1;2)$ *by* $k=\frac{1}{3}$.

   ANSWER $(3;\frac{8}{3})$

3. *Two similar triangles have areas 9 and 25. Find the ratio of their sides.*

   ANSWER $3:5$

4. *A rectangle 4 by 6 is enlarged by* $k=2.5$. *Find its new area.*

   ANSWER $24\cdot 6.25=150$

5. *Enlarge* $(2;6)$ *from the origin by* $k=-1$.

   ANSWER $(-2;-6)$

6. *A photo of area 40 cm² is enlarged so its area is 250 cm². Find k.*

   ANSWER $k=2.5$

7. *Enlarge the triangle* $(0;0)$, $(2;0)$, $(0;2)$ *by* $k=3$ *from the origin, and find both areas.*

   ANSWER $(0;0)$, $(6;0)$, $(0;6)$; 2 and 18

Hard · stretch and reason

7 PROBLEMS

1. *A shape is enlarged from (2; 3) by $k = 2$, and (5; 7) maps to which point?*

   ANSWER (8; 11)

2. *(1; 2) maps to (7; 14) under an enlargement centred at the origin. Find k.*

   ANSWER $k = 7$

3. *(2; 3) maps to (8; 9) under an enlargement with $k = 3$. Find the centre.*

   ANSWER (−1; 0)

4. *Two similar solids have volumes 27 and 216. Find the ratio of their surface areas.*

   ANSWER $k = 2$, so surface areas are in the ratio 1 : 4

5. *A map is redrawn at half the scale. What happens to the areas shown on it?*

   ANSWER They become a quarter of what they were.

6. *A triangle of area 12 is enlarged so that its area is 108. Find k, and the new perimeter if the original was 18.*

   ANSWER $k = 3$, new perimeter 54

7. *Explain why an enlargement with a negative scale factor turns the shape through 180°.*

   ANSWER A negative k sends each point to the opposite side of the centre along the same line, which is exactly a half-turn about the centre combined with an enlargement by $|k|$.

07 Homework

Homework — 5 problems

Cambridge Stage 9, Unit 13.4. State the centre and the scale factor every time.

1. *Enlarge (3; 5) from the origin by $k = 4$.*
2. *Enlarge (6; 2) from (2; 0) by $k = 2$.*
3. *A shape of area 8 cm² is enlarged by $k = 3$. Find its new area.*
4. *Two similar triangles have areas 16 and 81. Find the ratio of their sides.*

5. *(3; 4) maps to (9; 12) under an enlargement centred at the origin. Find k.*